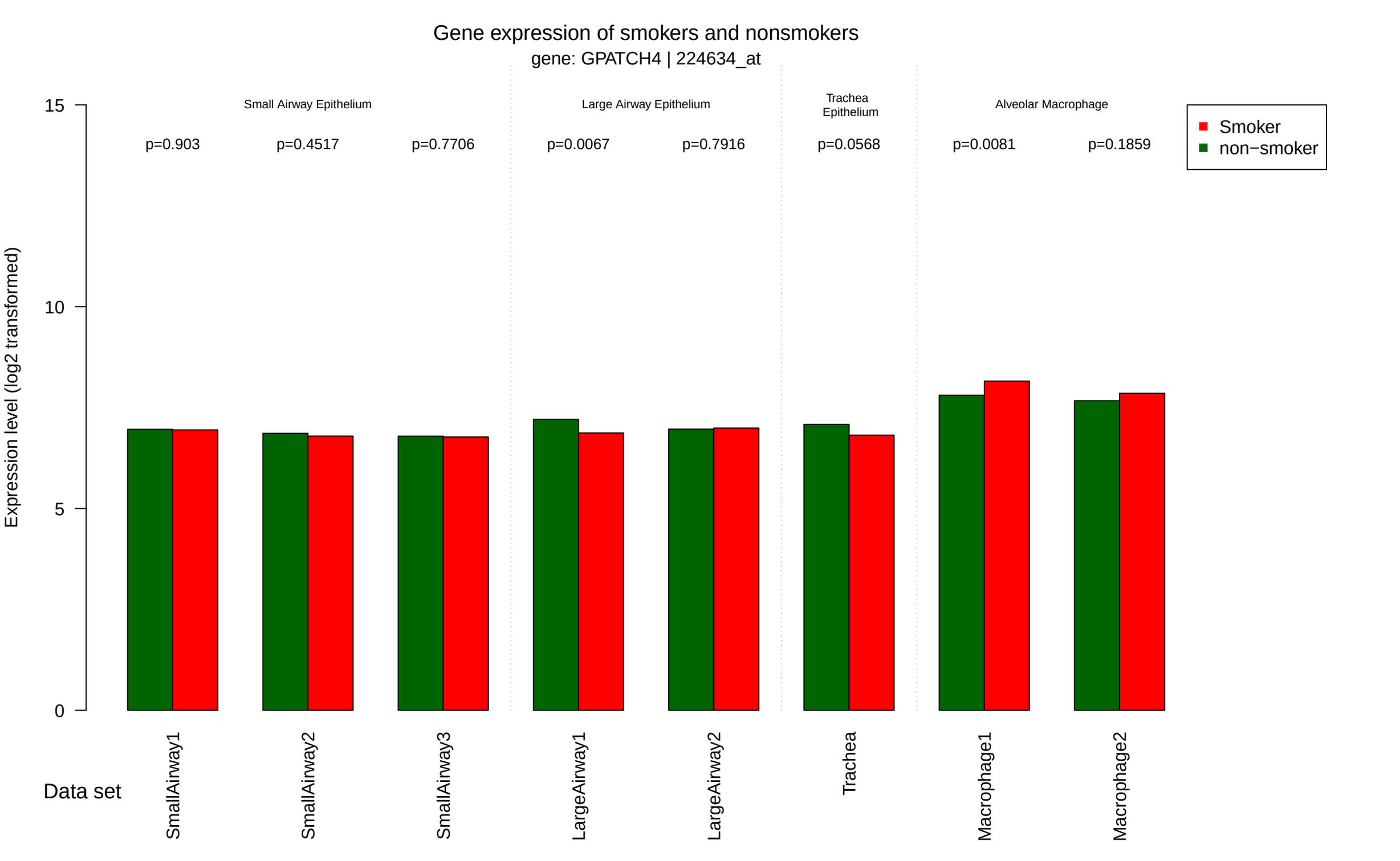
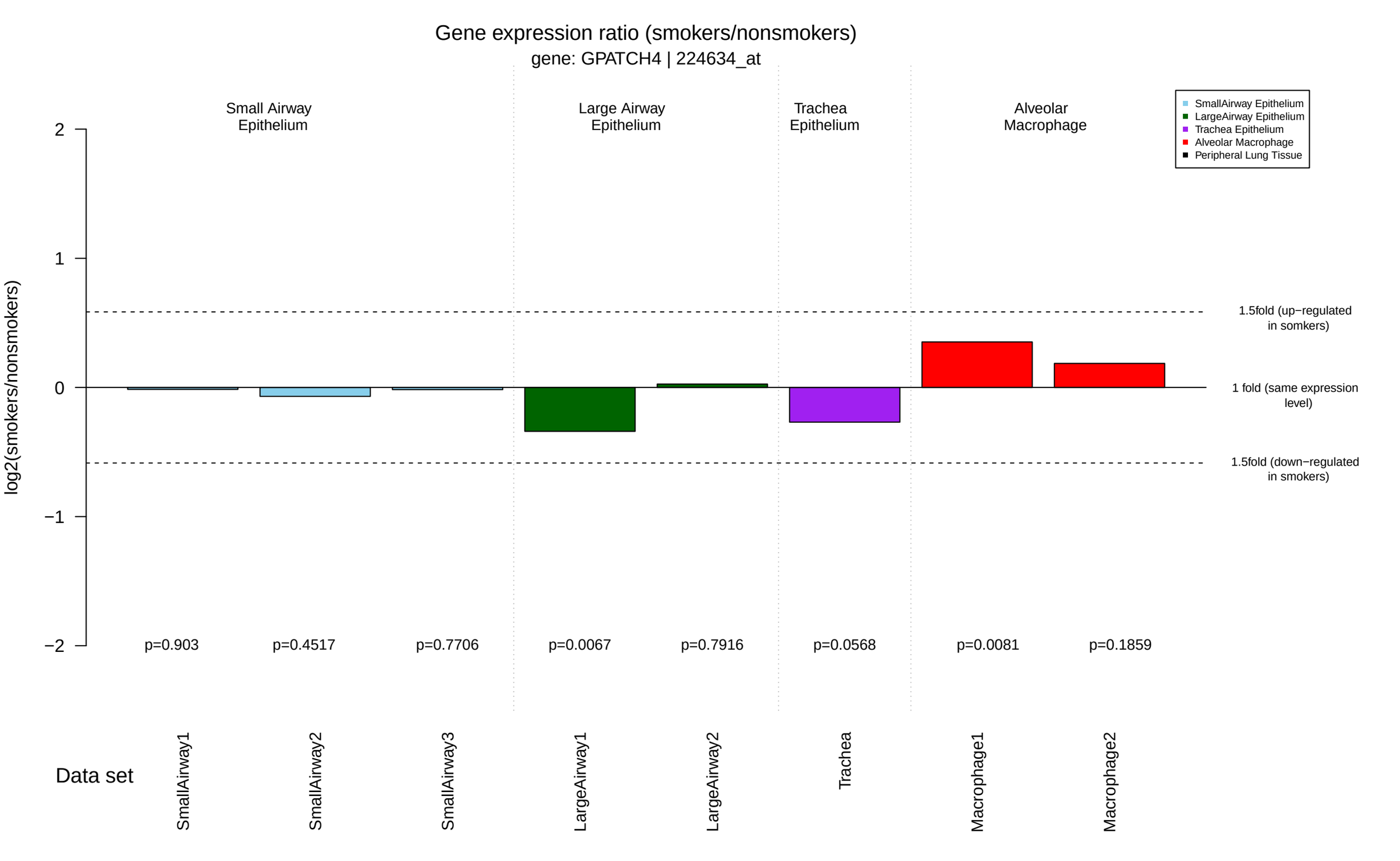
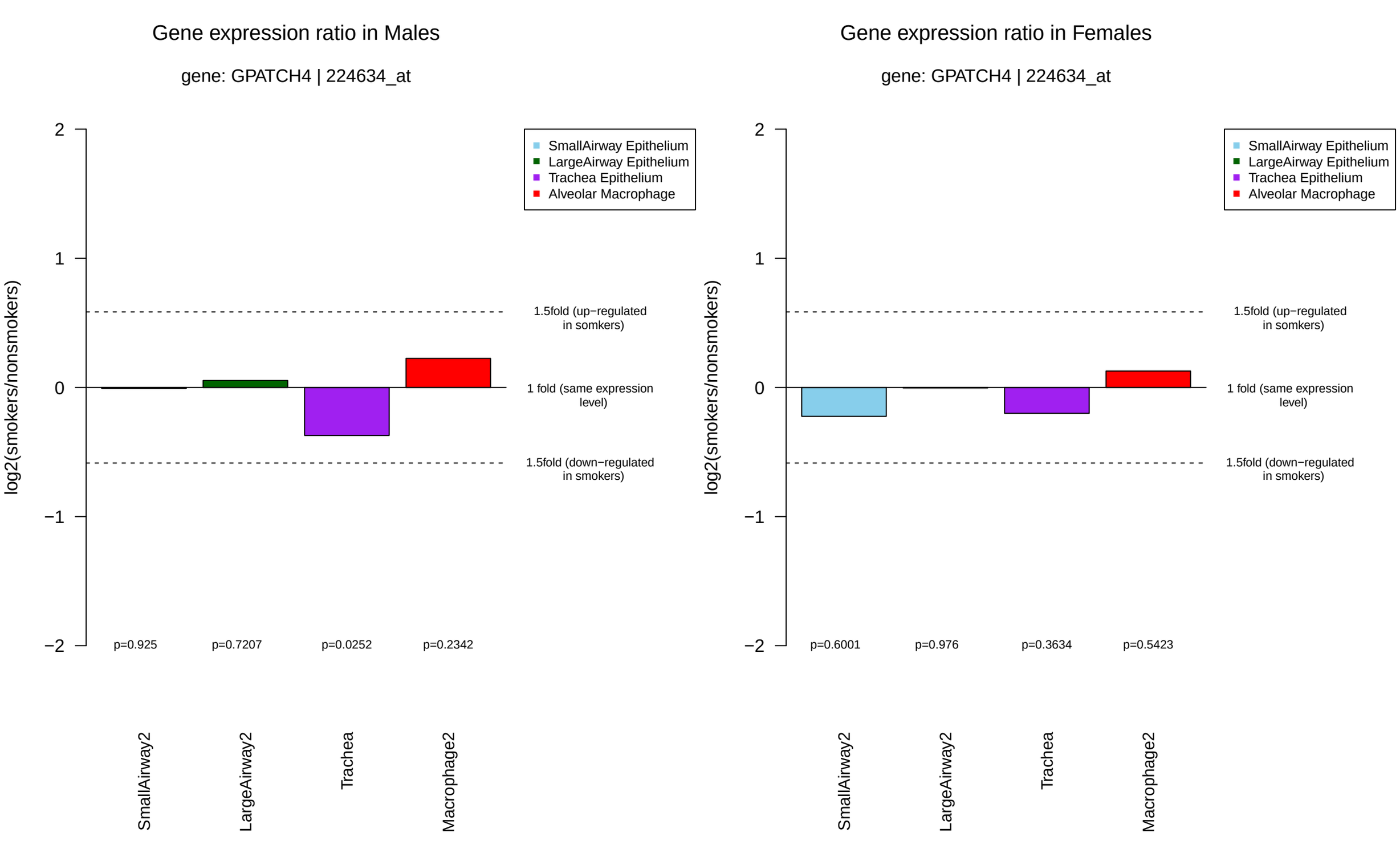
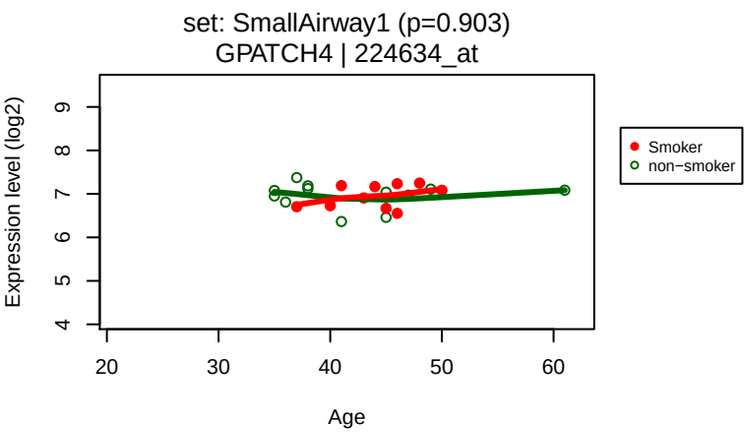
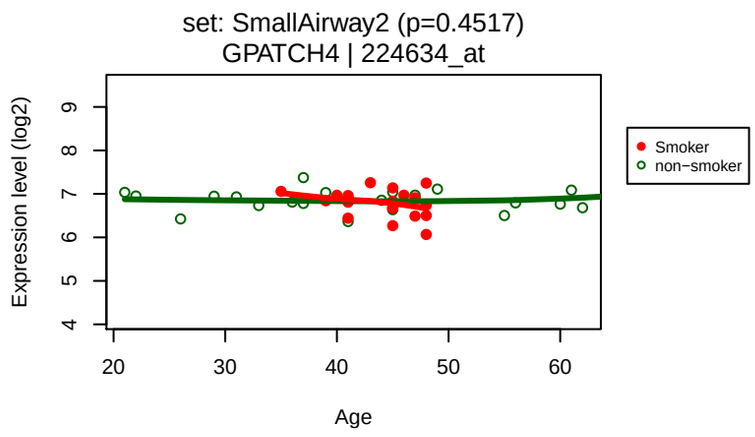
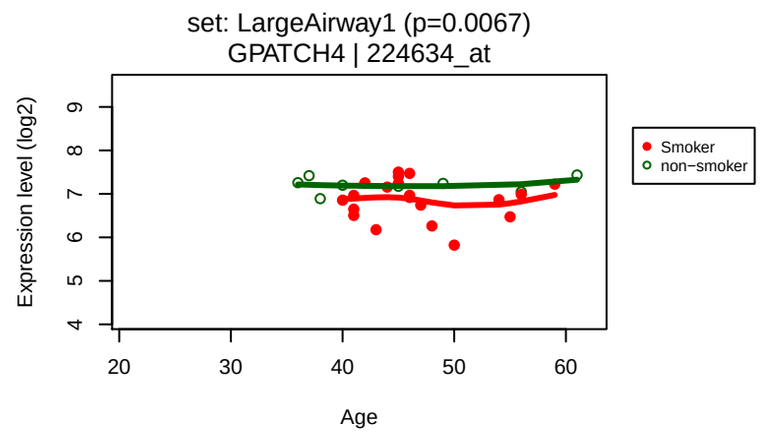

Gene expression of smokers and nonsmokers
gene: GPATCH4 | 224634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.903
p=0.4517
p=0.7706
p=0.0067
p=0.7916
p=0.0568
p=0.0081
p=0.1859
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

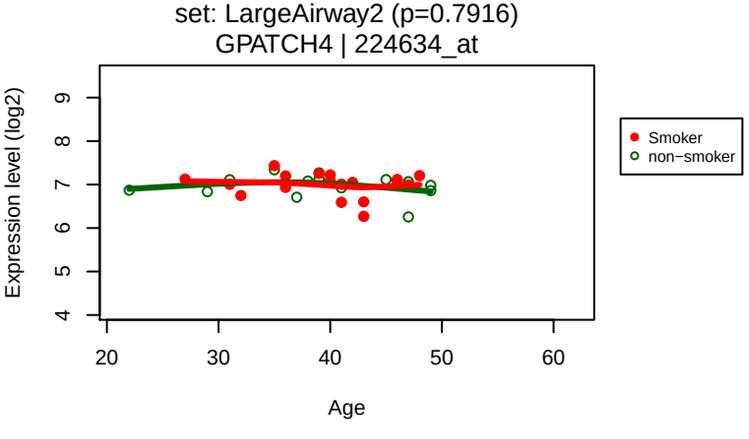

Gene expression ratio (smokers/nonsmokers)
gene: GPATCH4 | 224634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.903
p=0.4517
p=0.7706
p=0.0067
p=0.7916
p=0.0568
p=0.0081
p=0.1859
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

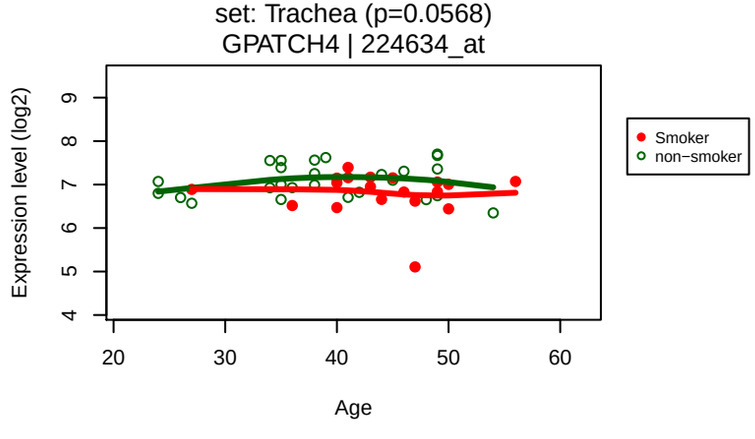

Gene expression ratio in Males
gene: GPATCH4 | 224634_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.925
p=0.7207
p=0.0252
p=0.2342
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPATCH4 | 224634_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6001
p=0.976
p=0.3634
p=0.5423
SmallAirway2
LargeAirway2
Trachea
Macrophage2

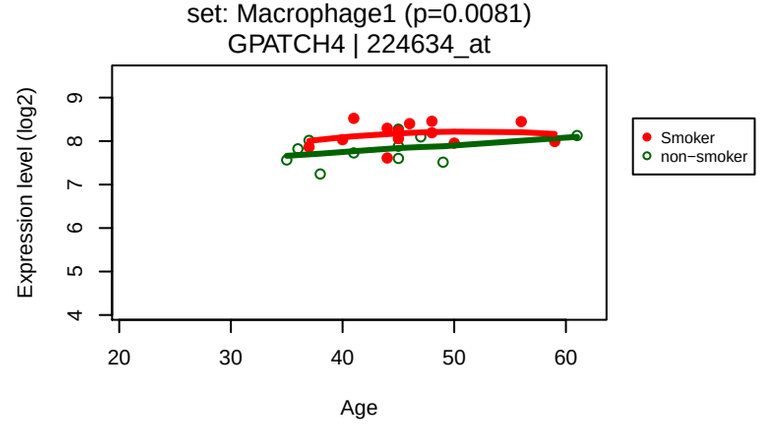

set: SmallAirway1 (p=0.903)
GPATCH4 | 224634_at
Expression level (log2)
Age
Smoker
non−smoker

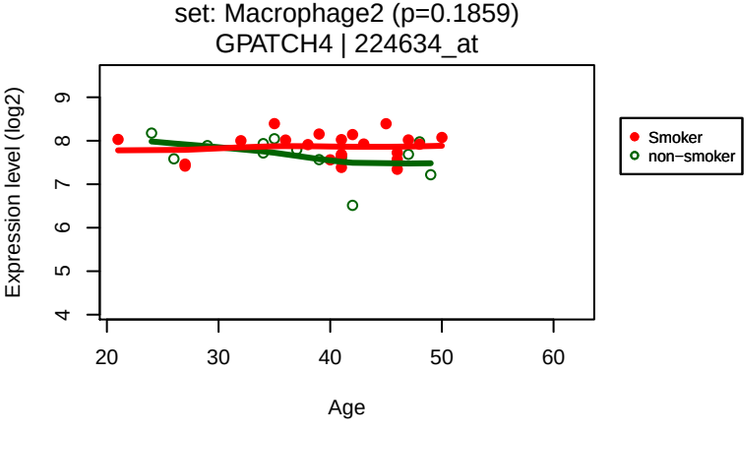

set: SmallAirway2 (p=0.4517)
GPATCH4 | 224634_at
Expression level (log2)
Age
Smoker
non−smoker

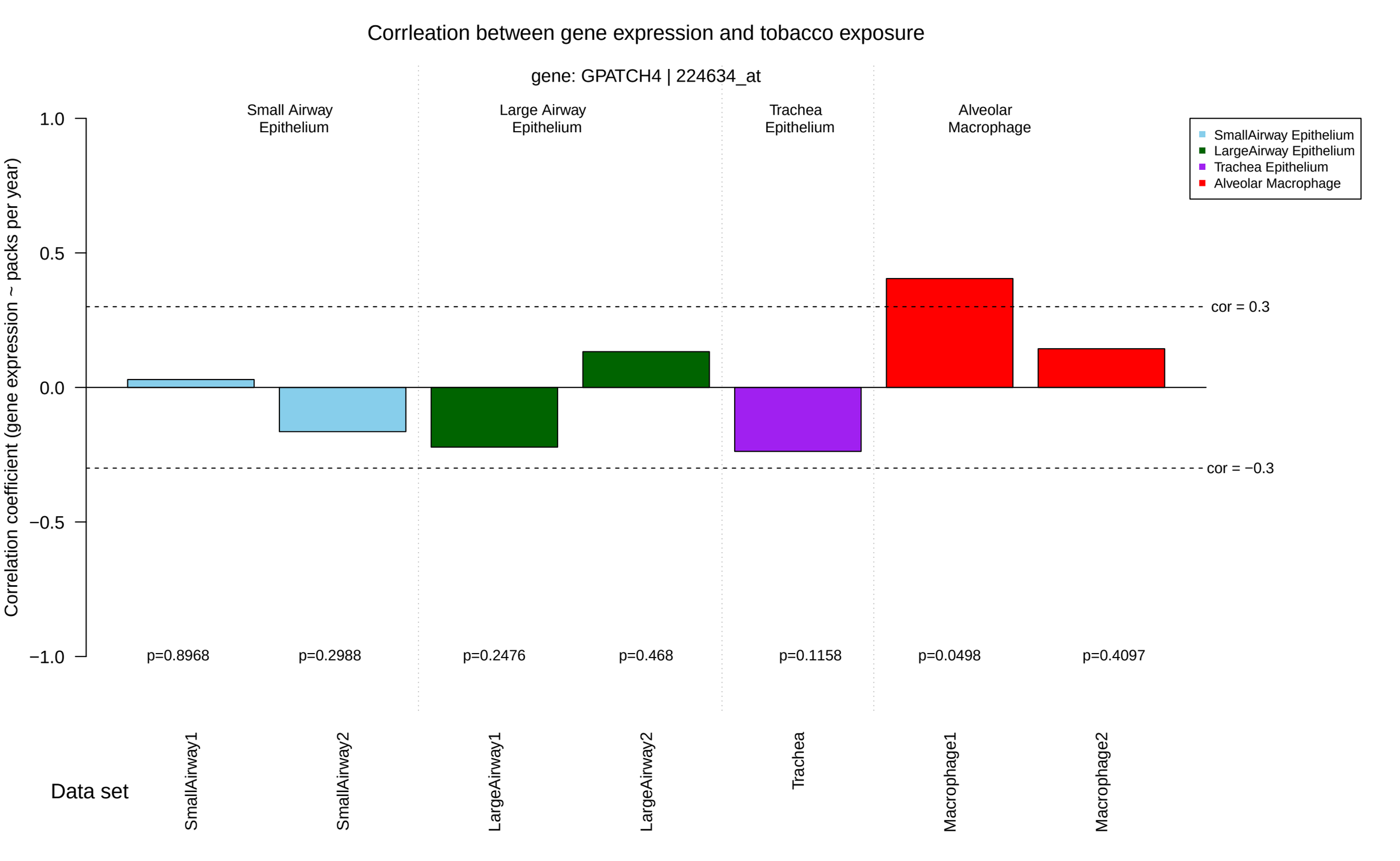

set: LargeAirway1 (p=0.0067)
GPATCH4 | 224634_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7916)
GPATCH4 | 224634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0568)
GPATCH4 | 224634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0081)
GPATCH4 | 224634_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1859)
GPATCH4 | 224634_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPATCH4 | 224634_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8968
p=0.2988
p=0.2476
p=0.468
p=0.1158
p=0.0498
p=0.4097
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2